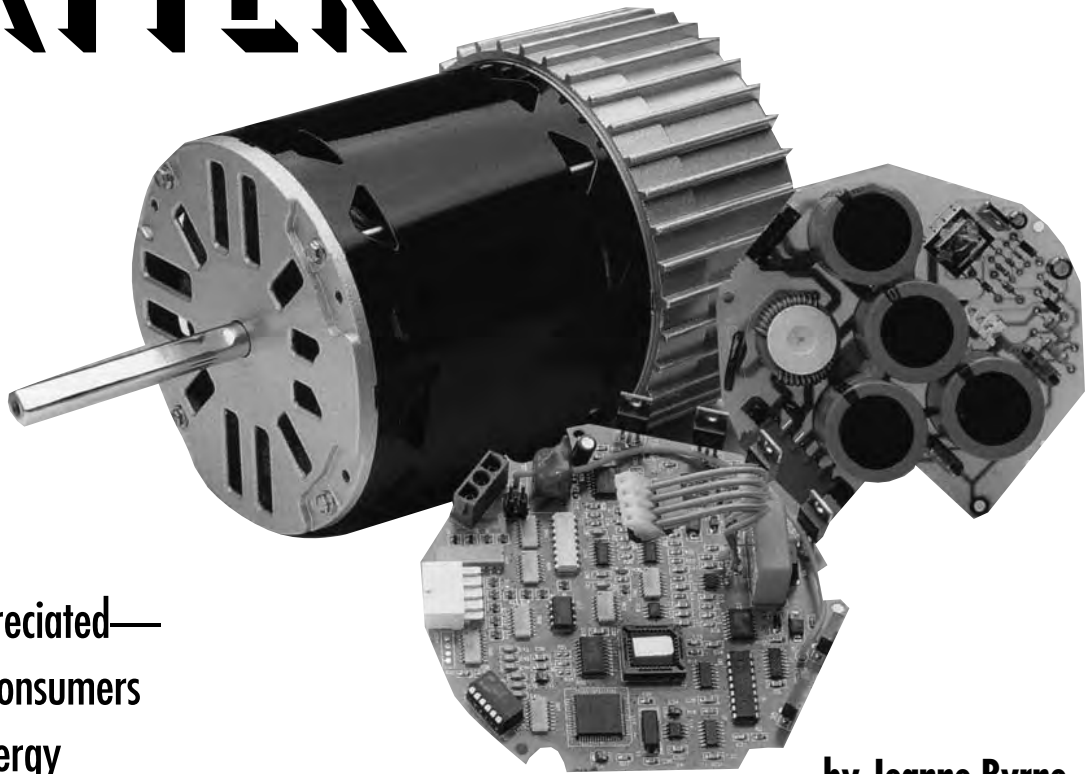


MOTORS MATTER



These underappreciated—
but growing—consumers
of household energy
deserve more attention.

by Jeanne Byrne

Apliance motors—the engines that drive air conditioners, refrigerators, washing machines, and other home appliances—account for about a quarter of all residential energy use. A new Department of Energy report estimates that in 1995, these motors—most of them less than 1 horsepower (hp)—used approximately 445 billion kWh, or about 4.9 quadrillion Btu of primary energy. Altogether, a home may contain dozens of motors, but consumers rarely think of motors when they look for ways to save household energy. This is because appliance manufacturers, who do not face the energy cost consequences of using inefficient motors in the home, purchase the overwhelming majority of motors for home appliances.



Improved motor efficiencies in newer refrigerators have yielded significant energy savings.

Appliance manufacturers do face appliance efficiency standards created by the federal government, however. Minimum appliance energy efficiency standards established under the National Appliance Energy Conservation Act (NAECA) in the early 1990s have led to improved motor efficiencies for many residential appliances—especially refrigerators and freezers. That is why replacing an older refrigerator or freezer can be such a great step for energy efficiency, (see “Monitoring Refrigerator Energy Usage” *HE* May/June ’00, p. 32). Currently, more than 84% of residential motor energy consumption occurs in appliances that are covered by NAECA. Although there are other ways to improve the overall efficiency of an appliance, using a better motor is often one of the most cost-effective ways for manufacturers to meet the standards.

More efficient motors are also often cost-effective for the consumer, although retrofitting existing appliances with more efficient motors is not practical in many cases. However, there are occasions when retrofitting an appliance with a new motor is necessary—when a furnace’s fan motor dies or when a new well or pool pump is needed, for instance.

Furnace fans can use quite a lot of energy. Replacing their motors with more efficient ones can bring real savings. Most forced-air furnaces and air

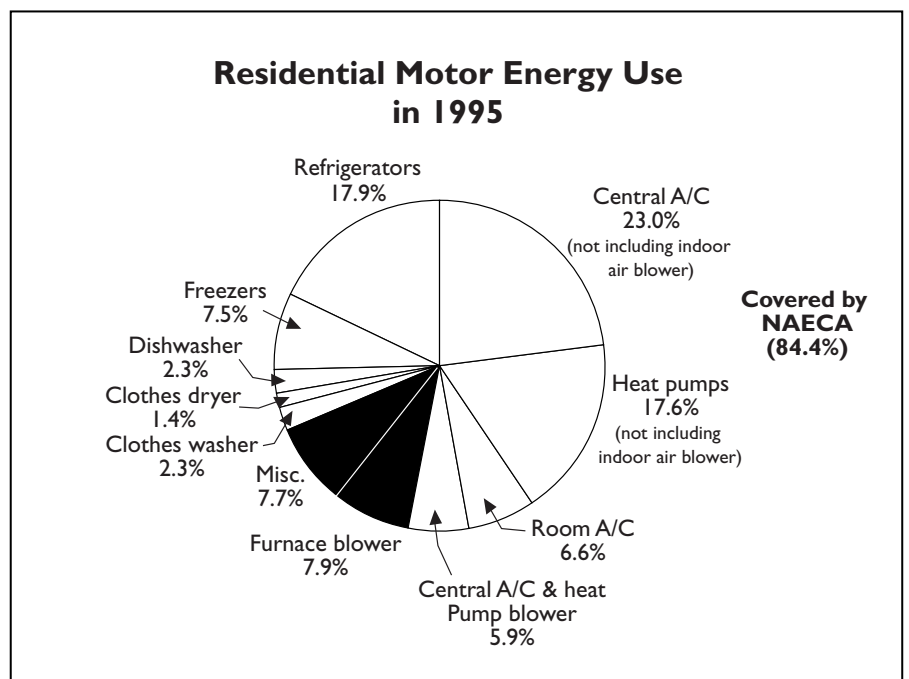


Figure 1. About 61% of residential motor energy is used in space conditioning. Another 25% is used in refrigerators and freezers. Washing and drying—laundry and dishes—uses 6%. The remaining 8% is used in small appliances such as vacuum cleaners and hair dryers; drills, saws, and other power tools; and other applications.

conditioners currently in use are powered by multiple-speed shaded-pole or permanent-split capacitor (PSC) induction motors, with efficiencies between 50% and 60%. One of the available speed levels is selected when the unit is installed, to match the blower output with the flow resistances in the space-conditioning equipment and the duct sys-

tem. Table 1 shows the potential energy savings for the average user from replacing the existing motor with each of three options: a high-efficiency PSC motor, an electronically commutated permanent magnet rotor (ECM), (see “Types of Motors,” p. 33), and a variable-speed ECM. Variable-speed motors can save the most energy and can be cost-effective

to the consumer if they are installed in high usage equipment. Recognizing the value of these savings, Canada’s new standards for the R-2000 program (the Canadian equivalent of Energy Star) will require the use of more efficient ECMs in furnace fans that are used in whole-house mechanical ventilation systems.

Room for Improvement

While NAECA has resulted in improved appliance motor efficiency, these motors are still not nearly as

Table 1. Efficient Motor Options for Indoor Blowers

Motor	Efficient Motor Option	Energy Savings Percent	Energy Savings \$/Year	Additional Retail Cost (\$)	Simple Payback
Central A/C blower	High-efficiency PSC	14%	\$4.50	\$15	3
	ECM	25%	\$8.00	\$40	5
	Variable-speed ECM	75%	\$24.00	\$75*	3
Heat pump blower	High-efficiency PSC	14%	\$11.20	\$15	1
	ECM	25%	\$20.00	\$40	2
	Variable-speed ECM	75%	\$60.00	\$75*	1
Furnace blower	High-efficiency PSC	14%	\$6.70	\$15	2
	ECM	25%	\$12.00	\$40	3
	Variable-speed ECM	75%	\$36.00	\$175**	5
Central A/C and furnace blower	High-efficiency PSC	14%	\$11.20	\$15	1
	ECM	25%	\$20.00	\$40	2
	Variable-speed ECM	75%	\$60.00	\$175**	2

*Cost of variable-speed blower only. ** Includes incremental cost of \$100 for capacity modulation in the furnace.

efficient as they could be. In the DOE study, researchers examining motor energy use and efficiency found much room for improvement, particularly in the appliances that use the most motor energy in the residential sector: HVAC equipment, refrigerators, and freezers.

Table 2 shows the average efficiency of several common residential motors, their potential practical efficiency, and the energy savings and payback periods that would result from using the more efficient motor.

Air Conditioners

The efficiencies of central air conditioners and heat pumps have been steadily increasing since the 1970s. The NAECA standards have driven the latest increases in the efficiency levels of these appliances in the last decade, raising them from 10.5 to 11.5 energy efficiency ratio (EER). Any further increases in the efficiencies of these appliances may be hard to achieve through motor improvements alone. Compressor motor efficiencies are now within 2% to 3% of the practical limits, leaving little room for increases in simple efficiency. For example, an 80% efficient compressor driven by a 90% efficient motor would have an EER of 11.8, only slightly higher than current products. However, simply improving the efficiency of the motors themselves is not the only way to save on appliance motor energy use.

Most motors operate at a single speed, cycling on and off to meet a load, or controlling the output with valves and dampers. VSDs can greatly improve how efficiently air conditioners' motors operate in applications with highly variable loads (see Table 3). They save energy by matching motor speed to load. VSDs also extend motor and system life.

Manufacturers have been reluctant to take this approach to increasing motor efficiency, because switching to VSDs tends to be more involved than incrementally increasing the motor

Types of Motors

Electric motors convert electrical energy into mechanical energy. This energy is then used to drive a fan, a compressor, or another rotating or oscillating part. But not all motors are alike, and the different types have different applications and efficiencies.

Single-speed induction motors are currently used for most residential applications, from portable fans to refrigerant compressors. Three-phase and single-phase squirrel cage induction motors and shaded-pole induction motors are included in this category. There is significant variation and a wide range of efficiency both within and between types of single-speed induction motor. Shaded-pole induction motors tend to be the least efficient. Generally, motor efficiency decreases with horsepower, so the smallest motors are usually the least efficient.

Universal AC/DC motors are commonly used for intermittent-use, high-speed applications, such as vacuums, blenders, and portable power tools.

Two-speed induction motors can improve efficiency for refrigerators, air conditioners, heat pumps, and distribution fans. With a two-speed motor, frequent cycling of one-speed motors at full power could be replaced with long periods of operation at half speed. Residential central air conditioners—and

some furnace blower motors, clothes washers, and ceiling fans—currently use two speed compressors (with 1–5 hp motors). Although the technology is available, few two-speed induction motors are manufactured in smaller sizes.

Electronic variable-speed drives convert fixed-frequency and fixed-voltage input power to adjustable voltage and frequency output power, enabling an associated motor to run at a variable speed. The motor can be a three-phase squirrel cage induction motor (SCIM), an electronically commutated permanent magnet rotor (ECM), or a switched reluctance motor (SRM). Of these, ECMs (also called brushless DC motors) are the most efficient—about 5%–10% more efficient than SCIM used for variable-speed operation. SRMs have varying efficiencies; typically, they are less efficient than ECMs but more efficient than SCIMs.

ECMs are a good option for residential HVAC and refrigeration equipment fans, which require motors less than 1 hp. SCIMs are mainly used in larger-horsepower industrial applications. SRMs are an old technology that is being successfully linked with new variable-speed-drive technology in, for example, Maytag's Neptune direct-drive front-loading washing machine.

Table 2. Potential for Residential Energy Savings through Increased Motor Efficiency

Application	Total National Motor Energy Use 10 ⁹ kWh/yr	Current Motor Efficiency Savings	Practical Efficiency* (Years**)	Potential Energy			Typical Payback
				%	10 ⁹ kWh		
Refrigerators and freezers							
compressor	101	80%	82%–84%	4%	4.0		14
condenser fan	6	15%	65%	77%	4.6		6
evaporator fan***	6	15%	65%	128%	7.7		4
Central A/C and heat							
pump compressor	159	87%	90%	3%	5.5		16
outside unit fan	21	50%	70%	29%	6.1		6
Room A/C compressor	25	87%	90%	3%	0.8		13
Indoor A/C and heating blowers	61	60%	80%	25%	15.3		3
Clothes washer motor	10	65%	75%	13%	1.3		10

Source: Opportunities for Energy Savings in the Residential and Commercial Sectors with High Efficiency Electric Motors, U.S. Department of Energy. *Based on upgrading installed motor base to maximum practical efficiency levels. **Assuming average electric rate of \$0.08/kWh.

***Evaporator fan savings include reduction in compressor load.

efficiency. It often requires a redesign for the whole appliance. But for certain appliances, such as air conditioners, a VSD can meet the appliance's load much more efficiently.

The capacity of an air conditioner's motor is usually based on the anticipated maximum cooling load of the house. But most of a home's operating hours occur under much milder conditions, requiring only a fraction of the cooling capacity of the air conditioner. In a typical air conditioner with a single-speed compressor, the unit cycles on and off to maintain the conditioned space at a desired set temperature. Continuous operation modulated by a VSD would use the heat exchangers more efficiently, limit losses associated with on-off cycling, and reduce the indoor air-flow rate.

Using a continuously variable-speed compressor with an ECM indoor blower could improve an air conditioner's seasonal energy efficiency ratio (SEER) by 50% (for example from SEER 10 to SEER 15). Using a two-speed motor would yield only a 30%–40% SEER increase but would cost somewhat less.

Motors also drive the fans that blow air over the condenser and evaporator coils. Using more efficient motors to blow air over the condenser fan of an air conditioner or heat pump can provide additional savings with a relatively short payback time (typically six years).

Refrigerators and Freezers

Residential refrigerators and freezers can have up to three motors—not

counting the ice maker and defrost timer. The largest (typically 1/8–1/3 hp)



The motor efficiency of most clothes washers is currently 65%.

drives the refrigerant compressor. In frost-free units, two additional smaller motors drive fans that force air over the condenser and evaporator.

The compressor motor is typically a 115V AC single phase, two-pole induction motor. The current NAECA standards have led to the use of more efficient induction motors. Even more stringent standards will take effect in 2001, reducing allowable

refrigerator and freezer energy consumption by an average of 25% from current levels. Manufacturers are beginning to assess higher-cost ECMs, which would be used as either a high-efficiency constant speed motor or a variable speed motor. The federal Environmental Protection Agency (EPA) has estimated that using a variable-speed ECM-driven compressor in combination with variable-speed ECM evaporator and condenser fan motors would save one-quarter of the total motor energy use. EPA estimated that this would increase the retail price by \$75; energy savings should pay the consumer back for this increase in about eight years. Variable-speed compressors also provide the benefits of very quiet, steady-state operation and rapid cooling when warm food is placed in the refrigerator.

Replacing only the fan motors, not the compressor, with variable-speed ECMs should reduce the payback period to only five years. This shorter payback period is due partly to the fact that increasing the efficiency of the evaporator fan also reduces compressor energy. The evaporator fan and motor are located within the refrigerated cabinet, so their use adds to the refrigeration load.

NAECA Standards Not Universally Helpful

There are a few appliances for which NAECA standards have not made much difference. In clothes washers and dryers, for example, the motors consume

Table 3. Potential for Residential Energy Savings through Variable Speed Motors

Application	Total National Motor Energy Use 10 ⁹ kWh	Current Motor Efficiency	Practical Efficiency*	Energy Savings		Typical Pay back (Years)**
				%	10 ⁹ kWh	
Refrigerator/freezer compressor	101	80%	88%	20%	20.2	8
Central A/C and heat pump compressor***	159	87%	90%	35%	55.7	15–25
Room A/C compressor	25	87%	90%	10%	2.5	20+
Indoor heating and A/C blowers	61	60%	80%	75%	45.9	2–3

Source: Opportunities for Energy Savings in the Residential and Commercial Sectors with High Efficiency Electric Motors. *Based on upgrading installed motor base to maximum practical efficiency levels. **Assuming average electric rate of \$0.08/kWh. ***Using a 2-speed induction motor. (Somewhat higher energy savings are possible with a continuously variable-speed motor, but the payback period is longer.)

only a small part of the total energy input compared to the heat input for water and air heating.

Although indoor blowers that are integral to air conditioner and heat pump systems are covered in the NAECA standards, these standards have not affected gas furnace indoor blowers. This is because the DOE gas furnace test procedure does not include the indoor blower power in the annualized fuel utilization efficiency measurement.

Appliance standards have gone a long way toward improving motor efficiency. But as motor-using appliances and gadgets continue to proliferate in American households, further efficiency improvements are necessary to reduce energy use. ■

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For more information:

“Opportunities for Energy Savings in the Residential and Commercial Sectors with High Efficiency Electric Motors” was prepared by Arthur D. Little for U.S. Department of Energy’s Office of Building Technology, State and Community Programs, December 1999. It is available from the National Technical Information Service (NTIS #PB2000-101159), U.S. Department of Commerce, 5285 Port Royal Rd., Springfield, VA 22161; Tel: (707) 487-4650. It can be downloaded from the Web at www.eren.doe.gov/buildings/documents/.